

# Deconstruction and Comparative Analysis of Large Language Model Hallucinations

## Introduction: Deconstructing AI Hallucinations: Insights from the HLE Benchmark

On the road to Artificial General Intelligence (AGI), we face a major challenge: "hallucinations" in Large Language Models (LLMs). Simply put, a hallucination is when a model is "confidently talking nonsense." It produces sentences that look logical, use correct grammar, and sound professional, but are actually wrong in terms of facts, reasoning, or instructions.

In the past, we used basic tests like MMLU to measure AI. However, top models like GPT-4o and Claude 3.5 have already mastered these exams. In 2026, the "Humanity's Last Exam" (HLE) was released as a much harder challenge. It collects expert-level academic problems that cannot be found on the internet. Results show that even the world's smartest models performed poorly, with accuracy rates between only 2.7% and 8.5%.

More concerning than failing the exam is the models' "blind confidence." Data shows a calibration error of over 70%. This means when a model gives a completely wrong answer, it often acts very certain that it is correct. This "high-confidence error" reveals a systematic failure in how AI handles high-level logic. This report will analyze real cases

from the HLE exam, explain why hallucinations happen, compare AI with "error-free" systems like databases, and discuss how to make AI more honest and reliable.

### **HLE Case Analysis: Specific Manifestations of Hallucinations**

Based on the expert-level questions and evaluation data collected from the HLE exam, we found that AI hallucinations do not occur because "it doesn't know," but because it repeatedly attempts "over-completion" at the boundaries of its knowledge. This phenomenon is particularly prominent in the fields of linguistics, mathematics, and chemistry:

#### **2.1 Fictionalization of Factual Dimensions: Professional-Sounding "Nonsense"**

In the field of linguistics, the HLE includes a deep and difficult question regarding the Tiberian pronunciation tradition of Hebrew (for example, identifying closed syllables in Psalm 104:7). This belongs to highly niche, expert-level knowledge for which almost no ready-made analysis can be found online.

**Hallucination Manifestation:** When answering, the AI acts like an expert. It skillfully uses technical terms such as "Shewa" and even cites non-existent pronunciation rules or solemnly attributes these details to a certain non-existent medieval manuscript.

**Root Cause:** The model is not "searching" for facts but is performing "statistical stitching." It pieces together fragments of relevant keywords from its training data that belong to entirely different contexts, fabricating a lie that sounds extremely plausible.

## 2.2 Logical Collapse in Reasoning: Memorizing Formulas but Failing Calculations

In advanced mathematics, especially in questions involving " $\infty$ -categories" (Infinity-categories) that require extreme abstraction capabilities, the AI's logical loopholes are fully exposed.

**Hallucination Manifestation:** The AI can usually list basic definitions and complex isomorphism formulas with 100% accuracy, which demonstrates its powerful memory. However, once it enters the specific logic of symbolic calculation, it begins to "self-contradict." For example, it might acknowledge that "A" is a variable in the first step, but treat it as a constant during calculation in the second step.

**Root Cause:** This is a typical "logical hallucination." The AI is merely mimicking the tone of a math professor but lacks true "Deep Thinking" capabilities (i.e., System 2 thinking). It cannot maintain consistent logic throughout a long chain of reasoning.

## 2.3 Falsification in Generation: "Empiricism" of Force-Fitting Templates

In the case of chemical pericyclic reactions, the question requires determining whether a complex chemical reaction is conrotatory or disrotatory.

**Hallucination Manifestation:** The AI can often correctly identify the type of reaction, indicating it has grasped the general direction. However, when explaining the specific Woodward-Hoffmann rules, it confidently applies rules meant for a " $4n$  system" to a " $4n+2$  system." To make its explanation seem more credible, it even invents a fake rule on the spot, such as "thermal reactions inevitably lead to same-direction rotation."

**Root Cause:** This is a "rule generalization hallucination." When a model falls into details it is uncertain about, it lazily chooses to use the most common and general patterns to cover specialized rules in unique scenarios.

## **The Deep Mechanisms of Hallucinations**

Through the analysis of HLE cases, we can see that hallucinations in Large Language Models are not accidental "mistakes," but a "systemic inertia" under the current technical architecture. The main reasons can be summarized into the following three points:

### **3.1 The Mismatch Between Statistical Probability and Logical Truth**

The underlying logic of LLMs is "probability prediction." Simply put, when answering a difficult academic question from the HLE, it is not performing rigorous logical deduction in its "mind," but is calculating "what is the highest probability for the next word."

For a model, if it predicts that "conrotatory" is 1% more likely than "disrotatory" after a complex chemical formula, it will choose conrotatory. It pursues linguistic fluency—"looking like the right answer"—rather than the objective truth—"being the right answer." This mechanism leads to the "statistical stitching" phenomenon: it pieces together relevant-looking academic terms perfectly, but the underlying chain of causality is broken.

### **3.2 "Path Dependency" and Knowledge Boundaries in Training Data**

HLE questions are difficult because they are "non-retrievable." This precisely hits the weak spot of the models.

Existing models are trained on massive amounts of internet text. They are better at solving "consensus-based problems" that have appeared thousands of times online. When facing "isolated knowledge" from the HLE that has never been seen and requires expert-level deduction, the model falls into "path dependency." It subconsciously applies the most common, plausible-sounding templates to handle new problems. Like a student who only memorizes questions, when encountering a new problem type, the model tries to force-fit old formulas, eventually leading to a "incorrect migration" of rules.

### **3.3 The "Overconfidence Trap" in Reinforcement Learning**

The reason models produce a calibration error as high as 70% (i.e., blind confidence) is largely related to their "tuning" process.

In the later stages of training, humans teach the model how to converse through "Reinforcement Learning from Human Feedback" (RLHF). During this process, human evaluators tend to give higher scores to answers that are "detailed, professional, and firm in tone," while potentially giving lower scores to honest but simple "I don't know" responses. This reward mechanism unintentionally taught the model a "survival strategy": even if you don't know the answer, you must imitate an expert's tone to fabricate one. Because it is encouraged to act like a "smart assistant" during training, it gradually loses the ability to judge its own "knowledge boundaries."

### 3.4 Lack of a "System 2" Calculation Engine

Nobel laureate Daniel Kahneman divided human thinking into "System 1" (fast thinking) and "System 2" (slow thinking).

Current Large Models are essentially enhanced versions of "System 1"—automated, intuitive fast responses. They lack the "System 2" calculation power found in calculators or databases, making them unable to perform self-verification and multi-step backtracking before providing an answer. Facing math or logic problems that require multi-step abstract reasoning, they cannot provide certain, verifiable evidence for every step like a deterministic algorithm. Instead, they "glide" all the way to the finish line; if one symbol is wrong in the middle, everything follows in error.

### Comparison Analysis with Deterministic Computing Systems

To more clearly understand the AI's "high-confidence hallucination," we can compare Large Language Models with traditional "deterministic systems" (such as databases, physical simulators, or mathematical software). Through this "black and white" reference, we can see the root defects in the reliability of Large Models.

#### 4.1 Logical Foundation: Strict Rules vs. Fuzzy Probabilities

- **Deterministic Systems:** Follow hard "If-Then" logic. For example, when you query a SQL database or use mathematical software (like Wolfram Alpha), the system runs according to preset code instructions.  $1+1$  always equals 2; if conditions are not met, it reports an error rather than guessing.

- **Large Language Models:** Run in a "probability field." Even when facing math problems, they are essentially predicting the "text sequence that looks most like the correct answer." This explains why AI can "memorize formulas but fail calculations"—it has mastered the "textual pattern" of the formula, rather than the "mathematical logic" behind it.

#### 4.2 Attitude Toward the "Unknown": "Null" Values vs. Creative Fill-ins

This is the most prominent characteristic of AI hallucinations.

- **Deterministic Systems:** Possess clear boundaries. If a piece of information is missing from the database, it returns a "Null" (empty value) or "404 Not Found." It would rather tell you "I don't know" than fabricate data.
- **Large Language Models:** Lack a "don't know" mechanism. In exams like HLE involving "non-retrievable knowledge," models often force-stitch relevant word fragments in an attempt to "complete" the conversation. They treat a lack of data as a "fill-in-the-blank" task, generating answers that sound highly professional but are actually entirely fictional.

#### 4.3 Process Transparency: White-Box Logic vs. Black-Box Mystery

- **Deterministic Systems:** Are "white boxes." Every line of code and every data flow is traceable and auditable. If a result is wrong, engineers can precisely locate which line of logic caused the error.

- **Large Language Models:** Are "black boxes." Hallucinations are often buried within the complex interactions of hundreds of billions of parameters. We cannot know for sure whether a fabricated Hebrew grammar rule stems from a fake paper in the training data or the abnormal activation of two unrelated neurons. This lack of traceability makes hallucinations extremely difficult to eradicate.

#### 4.4 Error Characteristics: Deterministic Failure vs. High-Confidence Deception

- **Deterministic Systems:** Failure is usually "explicit." Either the program crashes or it outputs garbled characters; users can tell at a glance that the system has malfunctioned.
- **Large Language Models:** Failure is usually "implicit." As HLE data shows, even when the model makes a mistake, its tone, format, and logical structure still maintain a high-level standard. This "high-confidence" error is more dangerous than an obvious malfunction because it significantly increases the cost of human verification.

#### Paths to Enhancing Model Reliability and Eliminating Hallucinations

Facing high-difficulty and high-confidence hallucination problems, the AI field is shifting from "purely pursuing model scale" to "pursuing reasoning rigor and honesty."

Below are several major improvement paths:

##### 5.1 From "Fast Thinking" to "Slow Thinking": Strengthening Chain-of-Thought

##### Reasoning

To solve the System 2 deficiency mentioned in Section 3.4, future models need stronger capabilities for "self-play" and "process verification." Like the OpenAI o-series models, they perform long-chain latent reasoning (Chain of Thought) before outputting an answer. Before forming a final conclusion, the model performs multi-step internal calculations and conducts self-checks on intermediate steps. This mechanism can effectively reduce logical collapses. If the model discovers a self-contradiction in the third step of reasoning, it can backtrack and correct it in time, rather than following an error all the way to the end.

## **5.2 Combining Symbolic and Neural Systems: Introducing External "Truth Checkers"**

Since large models are prone to straying in probability prediction, they should be equipped with the "deterministic systems" mentioned in Part IV as external plugins. This involves adopting a "neuro-symbolic" hybrid architecture. When encountering mathematical calculations, chemical simulations, or code writing, the model no longer dictates an answer directly. Instead, it generates a piece of executable code or a formula and calls external Python environments, Wolfram Alpha, or databases for computation. This transforms the model's role from an "encyclopedia" to a "dispatch center." Utilizing the rigor of deterministic tools hedges against the ambiguity of probabilistic models, fundamentally eliminating calculation-based hallucinations.

## **5.3 Honesty Calibration: Rewarding "I Don't Know"**

To address the issue mentioned in Section 3.3 regarding fabricating lies to please humans, we need to redesign the model's reward mechanisms. In the Reinforcement Learning from Human Feedback (RLHF) phase, the evaluation criteria must be changed. We should no longer purely reward "answering beautifully" but instead add an "honesty score." If a model chooses to honestly answer "I don't know" or "I'm not sure" when facing questions beyond its knowledge boundaries, it should be given a high-score reward. This aims to reduce that 70% calibration error. By "training" the model to recognize its own knowledge boundaries in this manner, the AI's confidence can be made to match its actual capabilities.